Why Your Grandmother Forgot Your Name

A Radio Engineer's Guide to Alzheimer's Disease

Every number from published meta-analyses.
No extraordinary claims. Just signal processing applied to neurology.

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1. The Problem

Alzheimer's disease has no effective treatment. Anti-amyloid antibodies slow decline by ~27% over 18 months but don't stop it. Over $40 billion spent on amyloid research since 1992. Trial failure rate: 99.6%.

This document proposes that AD is not one disease with one cause. It is six simultaneous failure modes of the brain's timing and synchronization systems.

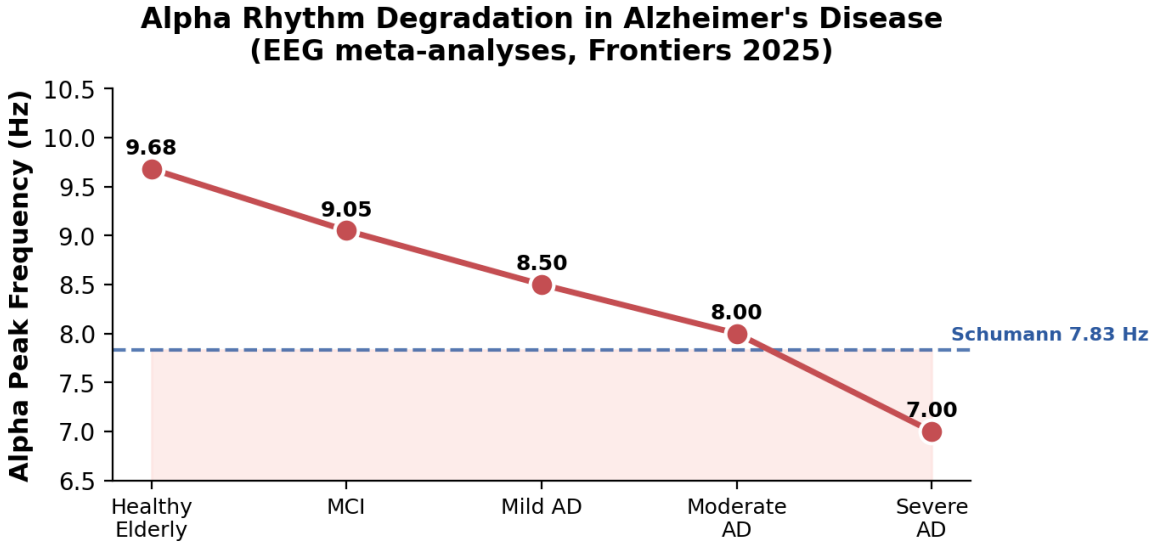

Figure 1. Progressive alpha degradation tracks disease severity

2. The Six Failure Modes

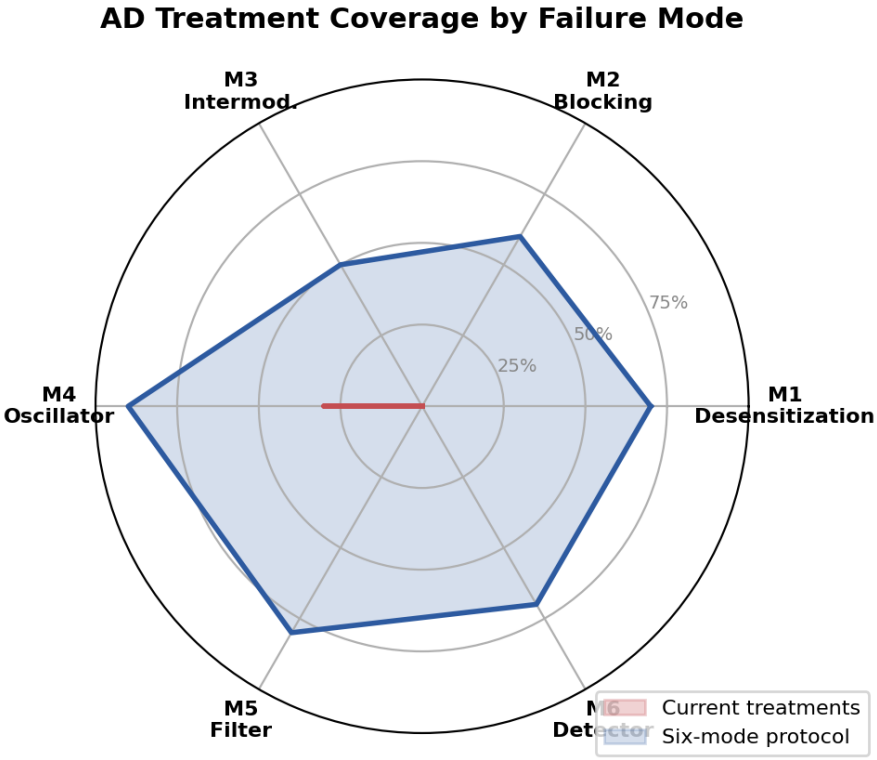

Figure 2. Current treatments address at most one mode

Mode	Radio Analogy	Brain Equivalent	Published Evidence
M1	Desensitization	Chronic EM exposure	RR 1.63 (meta-analysis)
M2	Blocking	Out-of-band saturation	AD deaths +140%
M3	Intermodulation	WiFi beacon 9.77 Hz	Calculated (novel)
M4	Oscillator drift	SCN neuron loss	42%, p<0.001 (Brain 2008)
M5	Filter degradation	Mg depletion	21 studies, p=0.045
M6	Detector failure	Complex IV reduced	30-70% (postmortem)

3. What Families Can Do Now

Free (covers 5 of 6 modes)

- Turn off WiFi at night ($15 timer) -- removes 9.77 Hz beacon during sleep
- Move bed away from wall -- reduces 50/60 Hz by ~75% at 1m
- Morning sunlight 30 min -- strongest circadian entrainment signal
- Daily walk 30 min -- most consistently supported AD risk reducer
- Consistent daily schedule -- external timing anchors for degrading clock

~$60/month (covers all 6 modes)

- Mg-L-threonate 2g/day ($30) -- 7.5-yr brain age reduction, N=100, p=0.043
- Selenium 200 mcg/day ($10) -- restores GPX4 substrate
- CoQ10 200mg/day ($20) -- supports mitochondrial electron transport
- Melatonin 0.5-3mg ($5) -- SCN receptor agonist

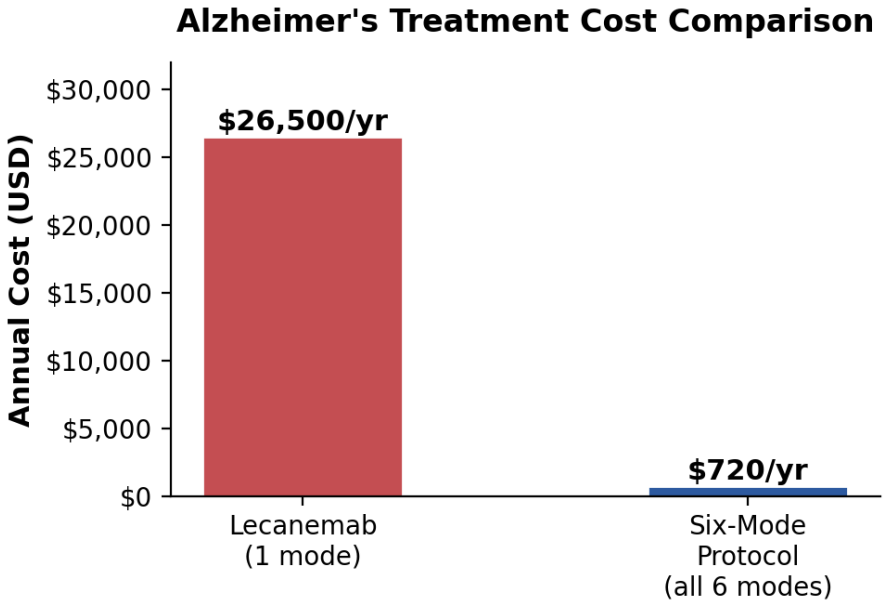

Figure 3. Annual cost: $720 (six-mode) vs $26,500 (lecanemab)

4. Published Sources

#	Source	Finding	DOI
1	ELF-AD meta (2017)	Occupational ELF: RR 1.63	10.1016/j.neuro.2017.09.007
2	Alz Assoc 2024	AD deaths +140% since 2000	10.1002/alz.13809
3	Frontiers 2025	Alpha 9.68->9.05 in MCI	10.3389/fnagi.2025.1522552
4	Frontiers 2021	Serum Mg depleted (21 studies)	10.3389/fnagi.2021.799824
5	MgT open trial	Improved FDG-PET + cognition	PMC6242385
6	Mito Complex IV	30-70% reduction (postmortem)	PMC7827030
7	GPX4/Se in AD	GPX4 reduced + ferroptosis	PMC2861544
8	SCN neuron loss	42% (p<0.001)	10.1093/brain/awn098
9	Light therapy	+1.7 hrs sleep	10.1371/journal.pone.0293977
10	Li prevention	RR 0.59 (41% risk reduction)	10.3389/fphar.2024.1408462
11	LATTICE trial	Full-dose Li failed in MCI	ALZFORUM
12	Braak 1991	Entorhinal first, cerebellum last	10.1007/BF00308809
13	Sex differences	Women 2x risk; XIST mechanism	Neuron 2024
14	AD-epilepsy	Bidirectional: 3.1x and 1.8x	10.1212/WNL.0000000000207423
15	Exercise + AD	Consistently reduces risk	Lancet Commission 2020
16	Lancet Commission	12 modifiable risk factors	10.1016/S0140-6736(20)30367-6
17	Mito timing	Early, before clinical Sx	10.1126/scitranslmed.abk1051
18	MgT RCT 2025	7.5yr brain age, N=100	10.3389/fnut.2025.1729164

For the grandchildren. For the grandparents. For everyone in between.

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